Status	Finished
Started	Tuesday, 20 October 2026, 1:39 PM
Completed	Tuesday, 20 October 2026, 1:50 PM
Duration	10 mins 52 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By comparing each element sequentially
c. By starting from the first element
d. By dividing the list length by two ✓

Question **2**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion
b. Bubble ✓
c. Selection
d. Complexity

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Linear search ✓
b. Binary search
c. Hashing
d. Dynamic search

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. Binary search algorithm is not efficient when the data elements more than 1500 ✓

d. Must use a sorted array

Question **5**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search

b. Sequential search

c. Linear search

d. Binary search ✓

Question **6**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops

b. The middle element is chosen randomly

c. The higher middle element is chosen as the middle element

d. The lower middle element is chosen as the middle element ✗

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers

b. It can only be applied to large lists

c. It is slower than linear search for small lists

d. It requires the list to be sorted ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is sorted ✓
c. When the list is unsorted
d. When the list contains duplicate elements

Question **9**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search backtracks to find duplicate elements
c. The search starts over from the beginning
d. The search stops immediately ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the element is found in the list ✓
c. When the list contains duplicate elements
d. When the list is sorted

Question **11**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 500
c. 100
d. 1000

Question **12**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. The first element is returned
c. The search is considered unsuccessful ✓
d. An error is raised

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)
b. O(n log n)
c. O(log n)
d. O👎 ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list
b. Deleting elements from a list
c. Sorting elements in a list
d. Determining whether an element is present in a list ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search can find multiple instances of the target element
c. Binary search works on unsorted lists
d. Binary search has a lower time complexity on large, sorted lists ✓